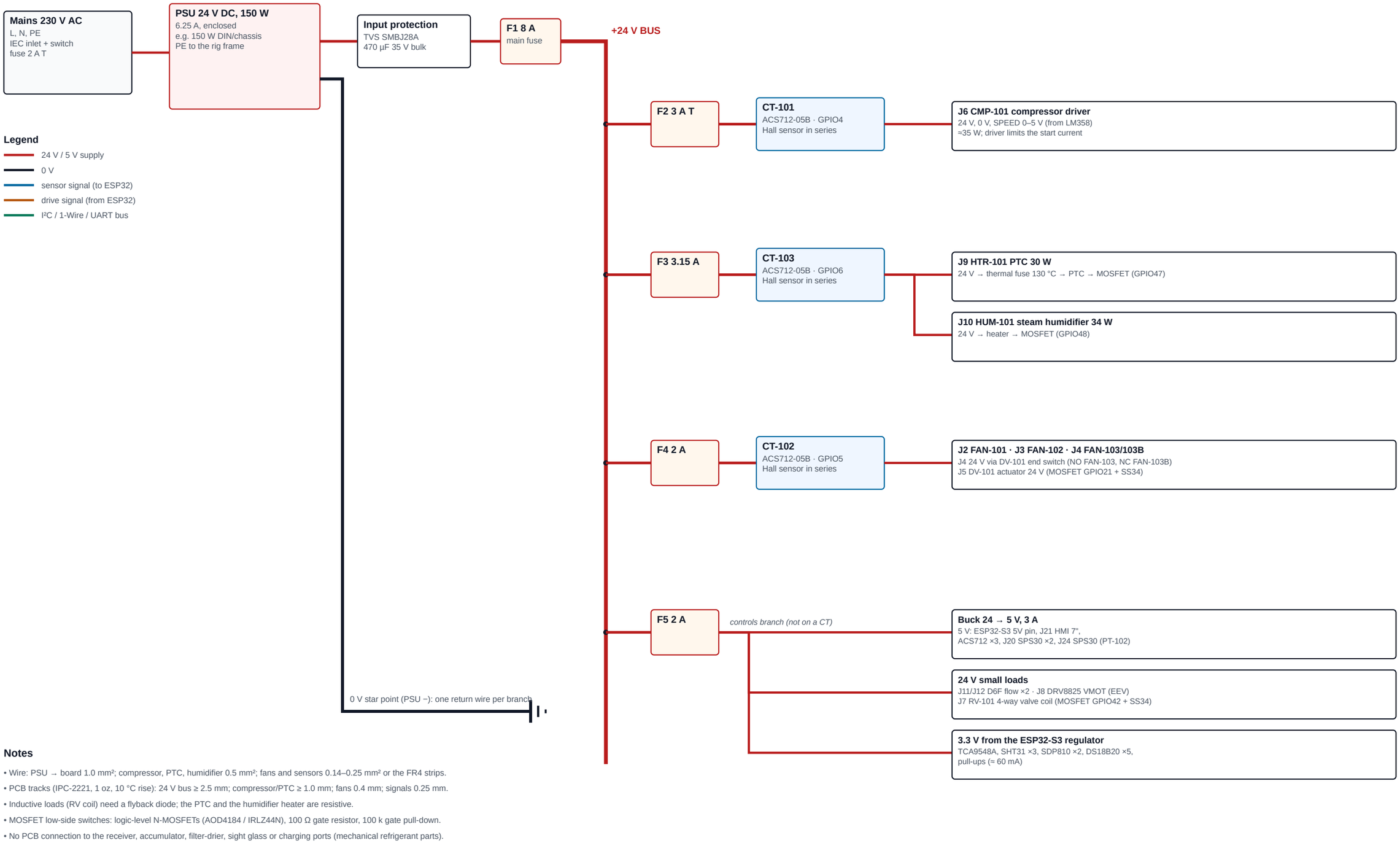
POWER DISTRIBUTION

All loads run from one 24 V DC supply. Each branch has its own fuse; the three ACS712 current sensors measure the compressor, fan and heater branches (CT-101 … 103). The controls branch is not on a CT (≈3.5 W, added in firmware).



Notes

- Wire: PSU → board 1.0 mm²; compressor, PTC, humidifier 0.5 mm²; fans and sensors 0.14–0.25 mm² or the FR4 strips.
- PCB tracks (IPC-2221, 1 oz, 10 °C rise): 24 V bus ≥ 2.5 mm; compressor/PTC ≥ 1.0 mm; fans 0.4 mm; signals 0.25 mm.
- Inductive loads (RV coil) need a flyback diode; the PTC and the humidifier heater are resistive.
- MOSFET low-side switches: logic-level N-MOSFETs (AOD4184 / IRLZ44N), 100 Ω gate resistor, 100 k gate pull-down.
- No PCB connection to the receiver, accumulator, filter-drier, sight glass or charging ports (mechanical refrigerant parts).

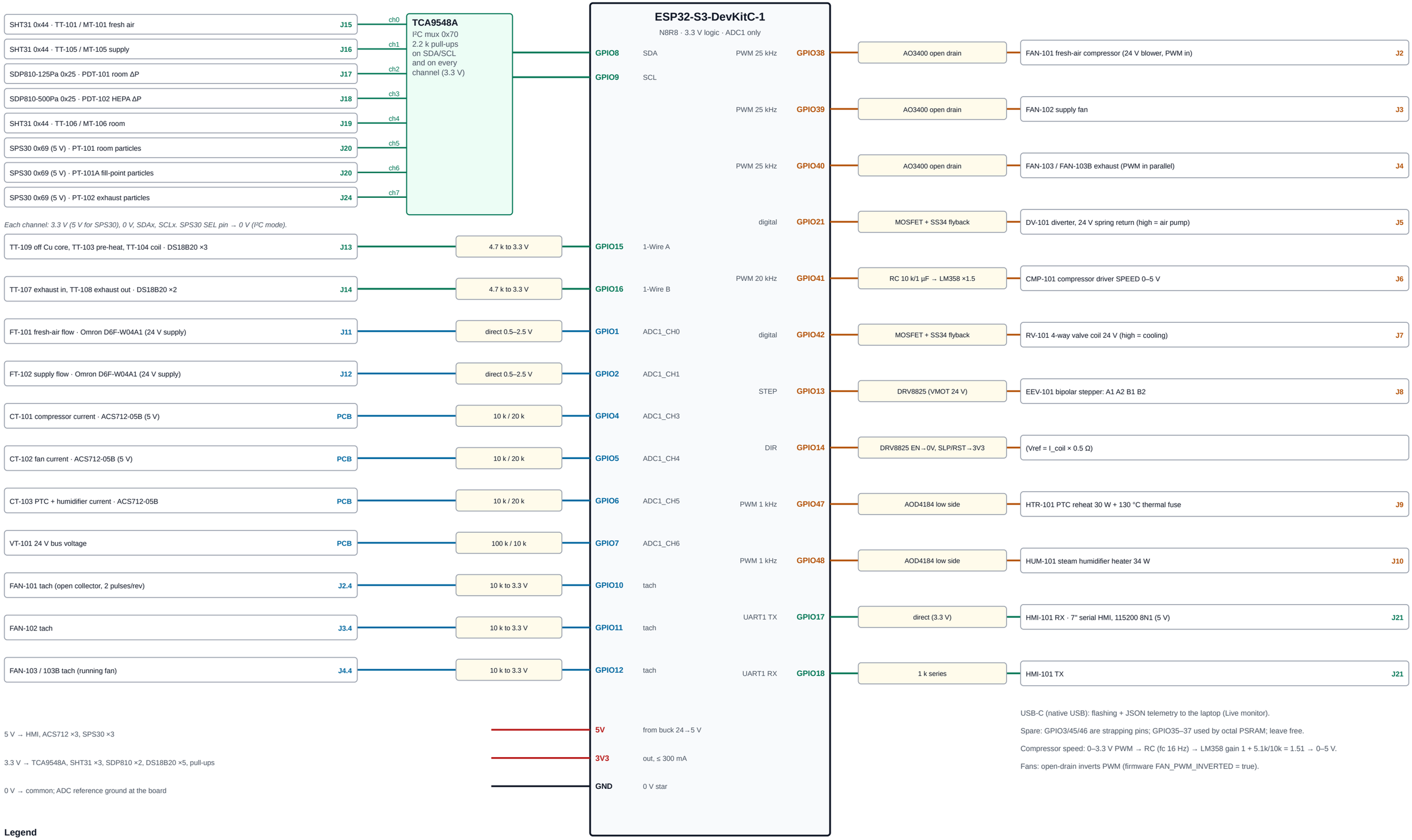
Clean-room heat-recovery air-pump rig

Sheet 1 · Power distribution, fusing and current sensing

ESP32-S3 controller · 24 V DC · pin map = firmware/include/config.h

CONTROLLER I/O

ESP32-S3-DevKitC-1 on the main PCB (back wall). Inputs on the left, outputs on the right. Every device is reached by its own FR4 strip with the connector shown (J-number). Interface parts sit on the main PCB.



Clean-room heat-recovery air-pump rig

Sheet 2 · Controller I/O: ESP32-S3 pins, interfaces, devices

ESP32-S3 controller · 24 V DC · pin map = firmware/include/config.h

Sheet 2 of 2 · A3 · Eng. Kariem Moussa · GIU